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Novel Particle Bee Optimisation based Framework for Fault Tolerance in Fog Environment

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Abstract. Fog Computing characteristics have fascinated its services in many applicability areas such as Industrial IoT, health monitoring systems and high performance computing, weather forecasting, agronomy, etc. Here, data is captured using the various sensor nodes installed at different locations and further processed by a cloud-based server to obtain the desired results. Heterogeneity of fog nodes and end devices in fog computing architecture is a major feature that helps to utilize already existing computing resources, yet it complicates the managing process. Such a diversified environment must be considered when designing fog environments in terms of computing capabilities, power consumption, and connectivity. In this work An efficient Novel Particle Bee Optimization based Fault Tolerance framework has been projected that ensures the implementation of Fog computing in real time. The proposed framework handling live server faults and also diminishes latency and bandwidth usage of the network and improves Quality of Service (QoS). The achieved results for the projected technique represent the reduction of network bandwidth and reduction in overall cost.

Keywords: Fog Computing, Particle Bee Optimization, QoS, Checkpoint, Replication, Fault tolerance.

1. Introduction

The next evolution of networks entails real-time reactions. Most IoT-based applications proliferate huge amounts of data by using the various sensors connected to things. This captured data may impose a lot of hurdles for any organization by exhausting the available network bandwidth [8], which in turn will require a further investment of time and money to overcome this issue. However, conventional cloud computing systems are not adequate to assemble the requirements of these new-era applications because of the geographical distribution of IoT devices and cloud based data centres. Therefore, there is a requirement for a novel computational model that expands the basic services across the network edge to the core of cloud computing, in order to decrease response time and transmission delays due to data movement from the edge to the cloud core [1]. There is a type of distributed computing called Fog computing that uses "edge" nodes to access cloud-like services. In fog computing, the term "Fog" means exactly the same thing as it does in everyday life. It is possible to encounter fog between clouds and the ground in the real world, and this concept is employed in fog computing to symbolise the region between clouds and the physical world[1][2].

The implementation of fog computing services along with cloud computing provides many benefits to clients and to organizations. Here, the various characteristics of Fog Computing include Quality of

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Service awareness, low latency, the reduction of bandwidth, avoiding unnecessary data transmission time, the decentralization of decision-making capability, and the interoperability of various communication technologies are some of the benefits. The managing of IoT in a fully mechanized environment examines the multifaceted system. Therefore, a module should be needed to become aware of the fault in the developed framework [2]. A variety of existing industries fraternity and associations are now accepting emerging IoT applications for getting better their performances. The emerging IoT technology also play a vital roles in the manufacturing industries for Monitoring and analysis of processing activities [12], Diagnosis and Detection of faults (MFDD) [2]. The workflow of the set of tasks assigned to the systems was monitored for ensuring the proper implementation of the proposed framework . Therefore, several effectual fault-tolerant and service placement frameworks must be proposed and implemented to address these issues.

This work implements the Checkpoints and replication schemes at the level of application and also at the system level. The elementary prerequisite is acquaintance regarding the occurrences of the collapse or failure in the system framework. The revival from that failure is commenced by restarting the job from the same point where it has halted or stopped functioning. Replication is the commonly engaged scheme for fault tolerance in fog, edge and cloud based environments in which failure occurs frequently. This scheme is generally engaged that improving the accessibility of the storage and computing capacities of distributed systems. Primary considerations in the replication scheme are an assortment of replicas and their placement because the storage system is usually complicated and broad [7]. The most important reason behind this work is to project and implement a proficient framework for fault tolerance and efficient service placement for fog-cloud based environment.

This work mainly projected a fog computing-based technique known as Fault Tolerance & Service Placement scheme. The proposed method combined the two strategies—Particle Swarm Optimization and Ant Bee Colony to create a novel method. The suggested method addresses live server fault tolerance as well by copying server operations and check pointing it when it's synchronized with a proxy server. Additionally, the system now performs better in terms of cost, network bandwidth use, and latency thanks to this work. The left behind sections of this document are prepared in a manner that: Section 2 reviews the various Fog-related techniques. Section 3 describes in detail the proposed method, "Fault Tolerance & Service Placement". Section 4 depicted and analyse the results of the anticipated scheme. Finally, Section 5 wrapping the projected work.

2. Related works

There are numerous research analysis have already been carried out that focuses on the various insights on the use of the fog based frameworks and its services. In [9], authors have presented the feasibility study of fog computing. In their study, they have considered the devices that generate a massive volume of data, frequency, and communication latency based on devices' geographical distances and analysed the Quality of Service (QoS) and the stakeholder’s experience. They have also highlighted the feasibility of computing in concern to online games for location awareness. The propose model improves the average response time by 20% with a reduction of 70% in data traffic between the remote server and end user device and also enhances the QoS.
In [6], authors have discussed the issue of reliability, detection and tolerance of faults in context to various IoT applications. They have considered the emerging smart city projects and focused on the utilizion of IoT and fog computing. In this work, they have considered the issue of fault tolerance, hardware malfunctioning, power loss, and cyber-attacks. The proposed model ensures that the fog nodes maintain the qualified operations and addresses the issue of operational disrupts and traffic problems. In another work, In [2], authors have focused on matching of MAC-GAC algorithms spatial and online assignment tasks. They have designed a mechanism for credibility and punishment using a combination of genetic and ant colony algorithm which is known as MQC-GAC. The task assignment

calculates the candidates for workspace based on the quality and assigns the worker credibility on completion. High-quality workers are motivated with rewards; else reduces remuneration. Thus MQC-GAC optimized the task assigned. In [3], authors have presented a segregated policy for optimization and placement of services in the fog computing based environment. The proposed approach addresses placing popular services nearer to the end user. The projected policy was implemented via iFogsim by implementing a micro service based requests with numerous experiments. The algorithm was executed on each device by considering the recital of the devices. The acquired outcomes was illustrated that their suggested decentralized optimization policy improves the overall recital of the system. In [5], authors have addressed , arranging the service replicas among the various fog nodes. An ant colony based optimization schemes has been proposed and applied to the various industrial scenarios. They have formulated the problem into two programmed goals, i.e. cost of processing and overall latency of service. The achieved results depicted that the projected method surpasses the baseline in terms of accuracy and diversity. In [7], authors have proposed and implemented a QoS based responsive scheme, which is the amalgamation of two well-known schemes known as Check pointing and Replication . The proposed scheme is one of the reactive fault tolerance techniques for effective detection of faults and another one is Bee-Mutation (BM) for proficient placement of the service. The monitoring mechanism of fog services has been employed to examine the fog node performance. The Check-points, Replication module and Bee Mutation module can also be accessed by the monitoring and analysis of services and also ensuring the projected framework was adaptive and fault-tolerant. In the comparative study of the projected IoT-FOG based framework with the CRBM, it was observed that the overall usage of network and cost of execution of the projected framework was little improved than the existing algorithm respectively. In [10], authors have proposed the QoS aware scheduling policy for efficient assignment of IoT based applications in the decentralized computing environments. Their proposed strategy could be set to maximizing the overall performance of the projected system, The proposed algorithm projected the Comprehensive Learning Particle Swarm Optimization based on Set (S-CLPSO) and also improves QoS parameters performance in terms of optimum usages of computational and network resources, makespan and budget satisfaction . In [4], authors have described various IoT applications scheduling schemes in Fog computing environments. They also recognized recent challenges and issues in fog frameworks i.e confront to real world challenges and their optimized solutions . In [11], authors have described the various QoE and QoS service placement and application placement schemes and policies in the real time and heterogeneous environment .

3. Proposed Methodology

The novel particle bee optimization frameworks handling faults at the server level in real-time environment, whenever the server is synchronized with a proxy server, replication and checkpointing operations are performed. These two exclusive strategies, Check Pointing, Replication and Novel Particle Bee Optimization, were derived from the original Ant Bee Colony and Particle Swarm Optimization scheme. Additionally, to develop fault tolerance in IoT-based fog computing, we have considered the following postulates:

i) The placement of the fog sensor devices allows for direct or indirect communication with several neighbour nodes.
ii) Each end user device make use of two fog nodes. Here, one fog node was supposed to be a main node, whereas the other one was considered as a backup node. The backup node perform its operation after the failure of the main node.
iii) A faulty node could only be revealed by its neighbor node, if any of the connected nodes cannot communicate.

The projected framework for fault tolerance and service placement as shown in figure 1 comprises three modules known as the frontend module, the backend module, and the communication module.

Here, the frontend module comprises various mobile devices. The front-end module gathers the data from the monitoring surroundings and sends it to the backend module for further necessary action through the communication module. The backend module represents the four steps, i.e., input population initialization, best cost optimization, fault tolerant sub module , and fog service allocation sub module. In the backend module, In order to optimize the best cost of fog node's ,the projected framework loads the input data , recognize the best source node and neighbor node in the environment. Finally, dispensation of the service is placed on the best available fog node deployed in the network.

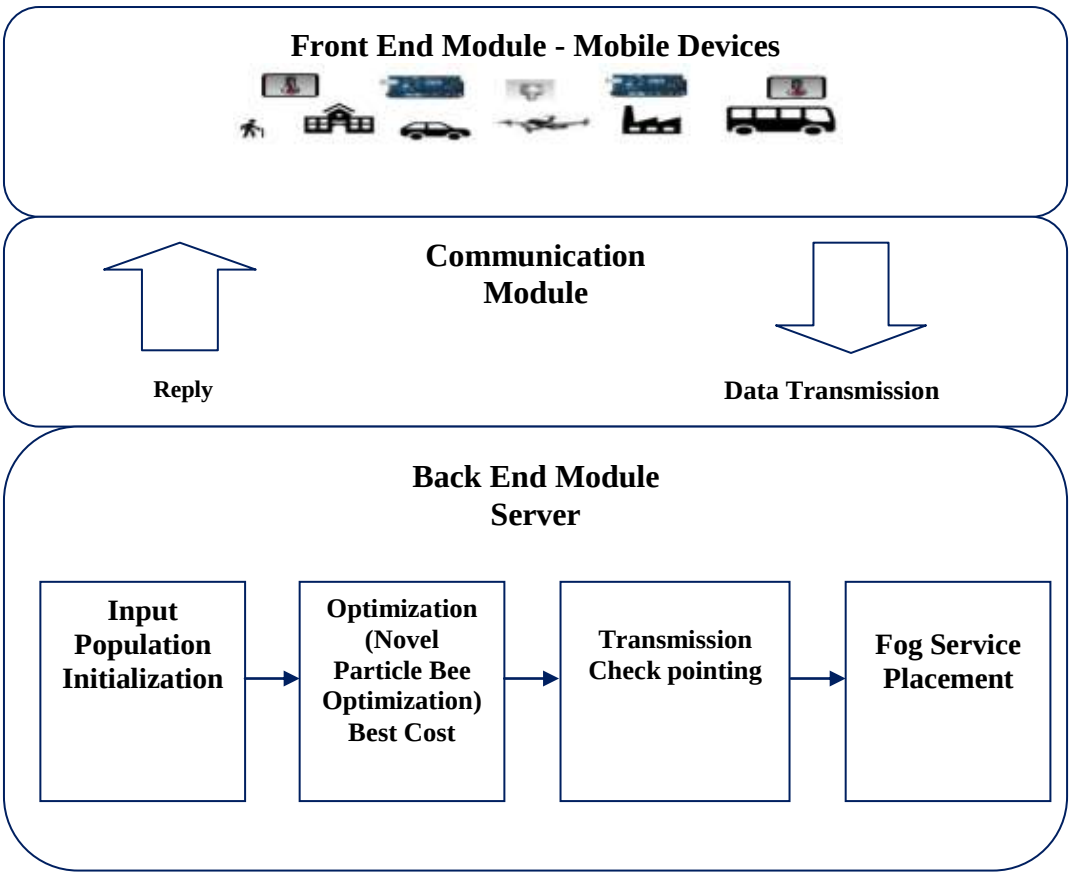


Figure 1.The framework of Fault Tolerance & Service Placement

3.1 Novel Particle Bee Optimization

Initializing the input as shown in the figure 1 and determining the origin node and neighbour nodes of the fog sensor environment are done via the projected scheme as depicted in the Algorithm I. Each transmission's completion time has been examined from a financial standpoint to determine the optimal course of action. The source of the fog sensor and its neighbour were connected at the lowest possible cost using several intermediate fog nodes. The device communication is facilitated through the centralized server. Computational and storage resources were concerted in the cloud data centers, and end-client devices can also be able to access the core cloud services. The below-mentioned algorithm represents the various function of the proposed system.

Algorithm 1- Novel Particle Bee Optimization

```
Begin
For(X==n; Y==m; Iterate++;){
Iterate()
            Until Solution(Best X, Best Y);
              }
             End For
             Initialize Input Population;
             Food Foraging of Bees()
             {
             Fog Sensor==Source;
             Calculate Source Cost;
             Fog Sensor==Neighbour;
Calculate Neighbour Cost;}
           Until Cycle for all sources completed;
          Display Best Solution Cost();
 End//Particle Bee Optimization
```

4 Results and Discussion

The mobile devices' transmission readings fixed with timestamps communicated from client to server are considered service placements. The proposed framework has been simulated using the extended version of iFogSim on the Windows operating system of which the configuration is Intel (R) Core(TM) of i7-4500U series CPU and 4G RAM PC. The proposed work implements a technique "Fault Tolerance & Service Placement" for the fog environment by using a games in virtual reality experiment as one of the case studies. This is divided into four data packets with a minimum turnaround time of 1 ms, a maximum turnaround time of 1 ms, and an average turnaround time of 1 ms. However, all packages are received with 0% packet loss. The check pointing and Replication technique's performance is evaluated based on the metrics response time, overall device throughput, scalability, performance, availability, usability, reliability, overhead, and cost for all the active devices. Throughput is the most crucial parameter for more effectual transmission of data, predominantly in the planned IoT-based fog framework. Throughput of the end client device has also been calculated using several packets transmitted by a particular device within a second. The response time was primarily considered as the total time requirement of the fog nodes to act in response to the received demand of the end client.

The primary obligation in this system is the fog node's availability to receive and progression of the upcoming demand in the uptime of fog node to the fog node's total time. Therefore, usability was calculated by identifying the number of storage and computational resources enthusiastic to culminate the process efficiently. On the other hand, the system's effectiveness might determine its dependability to produce precise results. In terms of the devices, Device ID 4 has the highest throughput (94%), response time (62%), scalability of the device (76%), availability of the device (96%), and reliability of the devices (85%). Figure 2 shows that Device ID 3 has the second-highest throughput of 88%, with

a response time of 79%, availability of the device 61%, device reliability of 97%, overhead related, and cost-effectiveness of 93 and 73, correspondingly .

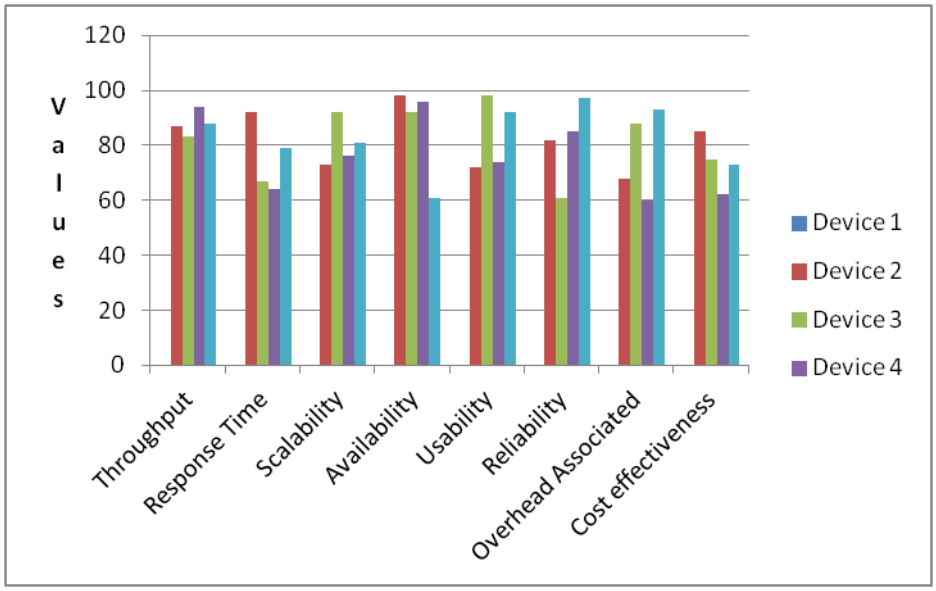


Figure 2. Fog Cloud Performance Statistics

Table 1 presents the Fog device execution delay for the game player state vr_game_0 and vr_game_1. As each device enables the game player simultaneously, the actuator signals can also perform the corresponding reply action for each transmission.

Table 1: Fog Device Tuple Execution Delay

Device	Execution Delay (sec.)
Player_Game_State	2.33
EGG	3.41
Concentration	0.18
Sensor	3.18
Global_Game_State	0.05

Figures 3 and 4 present the performance evaluation of the novel technique compared to the technique already existing Genetic Algorithm and Checkpointing, Replication and Bee Mutation. The comparison results show that Novel particle bee optimization leads to reduced usage of the network as 612680.3 Mbps and reduced cost of execution of IoT- fog as $ 81742.2. considering the other two existing algorithms.

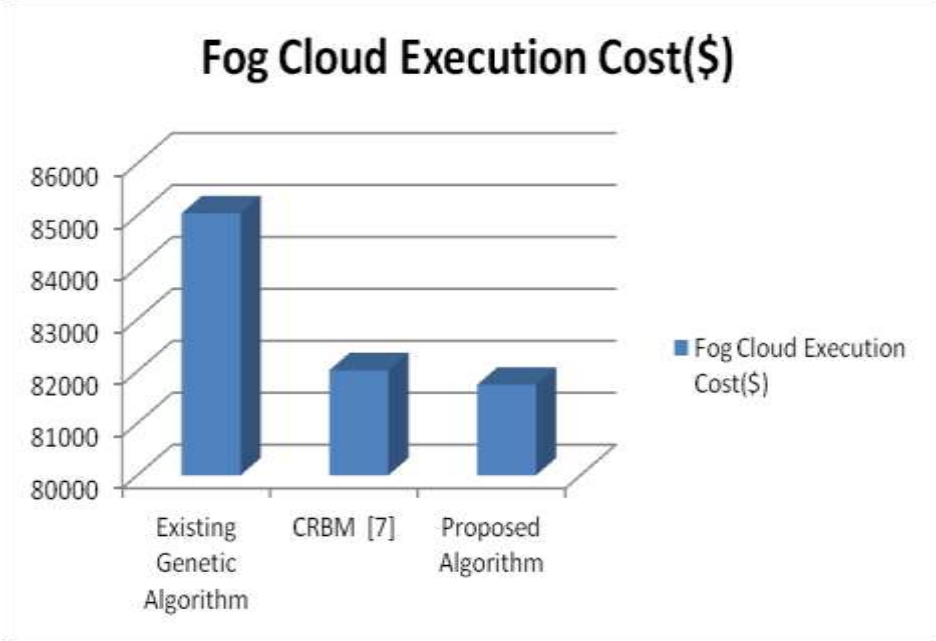


Figure 3. Comparison of Execution Cost.

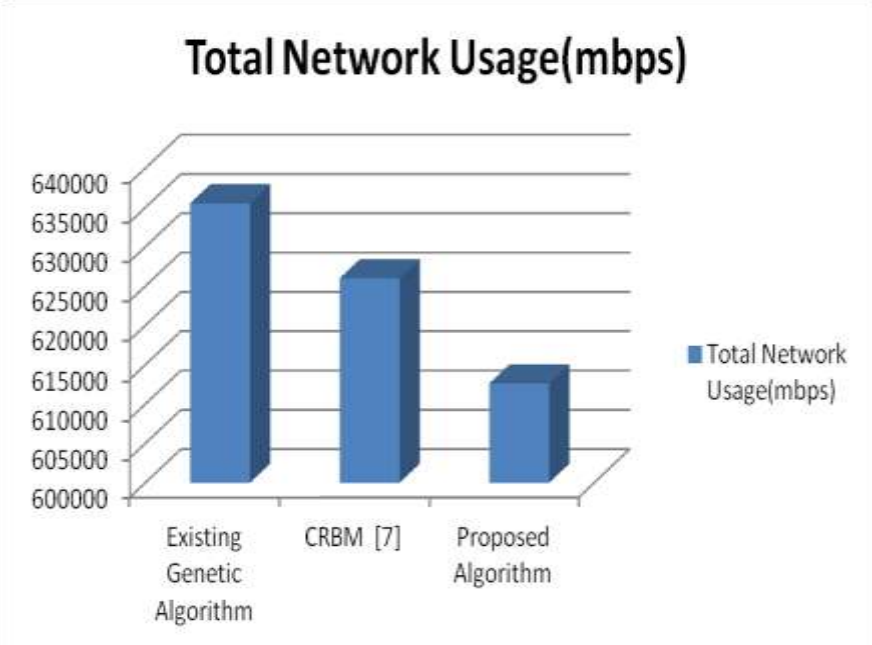


Figure 4. Comparison of network usage

5 Conclusion

The proposed framework "Fault Tolerance & Service Placement" addresses real time server faults via duplicating the server's operations by means of check pointing while synchronising it in the midst of a proxy server. Based on the results of the simulation , the projected framework minimises the usages of network up to 612680.3 Mbps for a fog cloud execution cost of $81742.2 also reduced in the context of evolving fog technologies. The devices also exhibit the maximum throughput (94%) for Device ID 4. We are planning to expand the projected framework in the applicability areas of Industrial IoT for proactive fault diagnosing , security and privace issues will also be looked in the future.

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